

Scope of Work (SOW)

Project Overview

The Local Redaction Service capstone project is an academic collaboration between **Virtual Gold Inc.** (the “Client”) and the Capstone Student Project Team, dated **October 5, 2026**. This version replaces the draft dated September 15, 2026. It reflects the direction agreed with the Client on September 21, 2026 and the design presented at the Midterm Presentation on October 5, 2026.

The project will design, build, and evaluate a **local redaction service**: a standalone HTTP service, running an open-source language model on project hardware, that replaces private information in executive email with reversible placeholders before the text reaches a cloud AI assistant, and restores the original values in the assistant’s reply. The Client’s own assistant calls the service automatically on every request.

The central challenge is determining what counts as private. The same string can be private in one context and public in another: a personal mobile number versus a published support line, or a name quoted inside a forwarded thread.

1. Project Objective

The primary objective of this project is to let an executive’s cloud AI assistant see only what it needs for the task, by detecting and redacting private data locally, automatically, and on every request, without a person reviewing each item.

The service will judge context rather than patterns alone, produce a calibrated confidence score for each item it finds, and make every redaction decision through rules in code, so that decisions are consistent and auditable. The guiding principle is that the model judges and the code decides.

The final outcome will be a working service, integrated with the Client’s VG08 assistant, together with a labelled email benchmark that measures how much private data reaches the cloud and what each part of the design contributes.

2. Scope of Work (In-Scope)

The student project team has completed an initial research phase covering sensitive-data classification, PII detection, local data minimization for cloud AI assistants, and confidence calibration of language models, including NIST SP 800-122, Microsoft Presidio, the Rescriber system (Zhou et al., CHI 2025), and studies of language-model overconfidence (Xiong et al., ICLR 2024).

The student project team will build the local redaction service. All detection and judgment run on project hardware with no outbound network access. The service will include:

- An HTTP API with request validation, through which the calling application submits text for redaction and later requests restoration.
- Segmentation of each email into body, quoted thread, and signature, with each segment tagged by its source.
- Candidate detection that combines three detectors (rules, Microsoft Presidio, and a local open-source model) and takes the union of their findings.
- Context assessment, in which the local model judges whether each candidate is private or public in its context and gives a score that is calibrated against labelled emails.
- A decision step in code that applies the sensitivity tier and threshold for each item.
- Replacement of redacted items with reversible placeholders (for example, [PHONE_1]). The mapping to the original values is stored locally and never leaves the service.
- A separate restore call that puts the original values back into the assistant's reply and flags any placeholder the service did not issue.
- A fail-safe path: if the model is unavailable or its output cannot be parsed, every candidate is redacted and the response is flagged as degraded.
- Request records that the caller can restore from or delete, and an audit log that records category, score, and action but never the original text.
- A monitoring dashboard (Prometheus and Grafana) showing requests, availability, latency per stage, the degraded rate, and model-call outcomes.

The student project team will define **sensitivity tiers and decision rules** based on NIST SP 800-122. The High tier covers identifiers, passwords and keys, bank and card numbers, health information, and business-confidential information. The Medium tier covers names of private individuals, personal phone numbers and email addresses, and home addresses. The Low tier covers cities, dates, organisation names, age, and gender, which are kept on their own and redacted when combined. Each tier has its own threshold; the higher the potential harm, the less confidence is required to redact. Thresholds start from initial values and will be tuned on measured results. The tier table will be provided for the Client's review.

The student project team will build an **evaluation benchmark** of 100 fictional executive emails, with every sensitive item labelled by type, exact position, and whether it must be masked. The benchmark includes context-dependent cases, clean emails that measure over-redaction, forwarded threads, emails containing prompt-injection attempts, and business-confidential statements that contain no single name.

The service will be evaluated on this benchmark. The primary metric is the **leak rate**: the share of private items that reach the cloud, reported per tier. Over-redaction, calibration, and processing time per email will also be measured. An ablation study, starting from a Rescriber-style baseline, will measure what each component adds (detector union, name propagation, repeated model runs, tier thresholds). Thresholds will be tuned on one subset of the benchmark and results reported on a held-out subset.

Testing will proceed in two phases. **Phase 1** tests the service on its own. **Phase 2** integrates it with the Client's VG08 assistant, which calls the service before any text is

sent to the cloud. The architecture will be tested with open-source models, cloud-hosted or local, on fictional data only.

The student project team will deliver a formal **Midterm Presentation** to the Client on October 5, 2026, covering the sensitive-data classification approach, the service architecture, the service engineering plan, the evaluation dataset, and the schedule.

The engagement will conclude with a formal **Final Presentation** demonstrating the completed service and its integration with VG08, the benchmark and ablation results, and recommendations. The student project team will additionally deliver technical documentation and a final academic report describing the methodology, architecture, results, limitations, and recommendations.

3. Out of Scope (Exclusions)

This project explicitly excludes deployment of the service into a live enterprise production environment or the commercial launch of a production product.

The service does not call any cloud AI model itself. The cloud assistant remains part of the Client's VG08 system and receives only the redacted text. Integration with VG08 is limited to the environment the Client makes available for Phase 2; the project will not integrate with Virtual Gold Inc.'s production systems, internal corporate databases, or other operational infrastructure unless separately approved and added to the project scope.

The project does not require access to or processing of real Personally Identifiable Information (PII), confidential customer information, or proprietary enterprise data. The benchmark and all testing use fictional data.

Per-item human review of redaction decisions is not part of the design; the service decides automatically on every request.

The vertical scenario discussed on September 21, 2026 (a trading assistant over public filings) is deferred and may be added only through a Change Request if time permits.

As an academic student project, the engagement does not include:

- Commercial software warranties or guarantees.
- Production-grade service-level agreements (SLAs).
- 24/7 system monitoring or technical support.
- Ongoing model retraining or continuous fine-tuning after project completion.
- Long-term maintenance of the service after the capstone concludes.
- Full production infrastructure deployment or enterprise-wide integration.
- Comprehensive certification against regulatory or cybersecurity standards.
- Commercial procurement or licensing negotiations with AI model providers.
- Large-scale production load testing requiring enterprise-scale infrastructure.

Any functionality or requirement substantially outside the objectives described in this document will require separate review and approval through the project change-management process.

4. Deliverables

The student project team will produce the following primary deliverables:

1. Local Redaction Service

A working local service with an HTTP API that performs the segmentation, detection, assessment, decision, placeholder, and restore steps described in Section 2, with the fail-safe path, request records, and audit log, and that is integrated with VG08 in Phase 2.

2. API Contract

A versioned specification of the service's endpoints, request and response formats, and error handling, frozen at the VG08 integration milestone.

3. Sensitivity Tiers and Decision Rules

The tier table, its categories and thresholds, and the rationale for each, provided for the Client's review and adjusted from measured results.

4. Evaluation Benchmark

100 fictional executive emails with every sensitive item labelled, in a shared format so that any model's output can be scored the same way.

5. Evaluation Report

Results on the benchmark, which may include:

- Leak rate per tier.
- Over-redaction rate.
- Calibration of confidence scores.
- Processing time per email and load-test results on the confirmed hardware.
- The ablation study showing what each component adds.

6. Monitoring Dashboard

A live dashboard of the service's health: requests and status codes, availability, end-to-end and per-stage latency, degraded rate and causes, model-call outcomes, record capacity, and the policy, model, and prompt versions in use.

7. Midterm Presentation

A formal presentation to Virtual Gold Inc. on October 5, 2026, covering the sensitive-data classification approach, the service architecture, the service engineering plan, the evaluation dataset, and the schedule.

8. Final Presentation

A formal presentation demonstrating the completed service, including:

- A demonstration of the service integrated with VG08.
- The architecture.
- Benchmark and ablation results.
- Key conclusions and recommendations.

9. Final Documentation

Supporting documentation describing the architecture, the API contract, the evaluation methodology, results, limitations, and recommendations, together with the final academic report.

5. Project Timeline

The schedule below follows the service engineering schedule presented at the Midterm Presentation (October 5 to November 30, 2026).

• **Research, Requirements, and Design** – *Weeks 1–6 (completed)*

Research, client requirements, the change of direction on September 21, the sensitivity tiers, the service architecture, and the evaluation benchmark.

The **Midterm Presentation** to the Client is scheduled for **October 5, 2026**. – *Week 7*

• **Contract and Service Skeleton** – *Week 7, October 5–11*

API contract version 0.4.2 and the service skeleton. Dashboard and load-testing work begins.

• **Fall Break** – *Week 8, October 12–18*

No project work.

• **Service Build** – *Weeks 9–10, October 19 – November 1*

Gateway, request records, and model adapter (week of October 19); pipeline wiring and failure path (October 19 – November 1).

• **Phase 1: Standalone Testing** – *Weeks 10–11, October 26 – November 8*

The redaction service is tested on its own. Dashboard and load testing in the week of November 2.

- **Phase 2: VG08 Integration** – *Weeks 11–12, November 2–15*

Integration with the Client's VG08 service, with hardening and the contract freeze in the week of November 9.

- **Documentation and Final Evaluation** – *Week 13, November 16–22*

Documentation and the final evaluation on the benchmark.

- **Team Materials and Rehearsal** – *Week 14, November 23–29*

Final presentation materials and rehearsal.

The **Final Presentation** to the Client is scheduled for **November 30, 2026**. – *Week 15*

Milestones:

- October 11 – Contract 0.4.2 and service skeleton.
- November 1 – End-to-end pipeline with failure path.
- November 8 – Dashboard and load-test results.
- November 15 – VG08 integration; contract freeze.
- November 22 – Code freeze (soft freeze November 18).
- November 30 – Final presentation.

6. Assumptions and Dependencies

The Client will provide **GPU and sandbox access** for the local model.

Where the project requires paid resources, such as GPU or other hardware resources, sandbox compute, or model API usage, the Client will sponsor or cover these costs. The student project team will agree any such usage with the Client in advance.

The Client will make **VG08** available for integration by November 2, 2026, when Phase 2 begins.

The Client will review the **tier table and term list**.

The Client will confirm whether the API request can label each text's source.

The project assumes that no Personally Identifiable Information (PII), confidential customer information, or proprietary enterprise information will be required to complete the agreed scope. All testing uses fictional data.

The project depends on continued availability of the selected open-source models, software frameworks, and development tools. Significant changes to model availability, licensing terms, or technical restrictions may require modifications to the implementation or evaluation approach.

The evaluation will be limited by the compute resources reasonably available to an academic student project. Consequently, testing may focus on representative workloads rather than production-scale enterprise traffic.

The Client will provide reasonable and timely feedback on major project milestones, architectural decisions, prototype demonstrations, and presentation materials to maintain the academic project schedule.

Any significant modification to the objectives, deliverables, technical scope, or agreed project responsibilities described in this document should be documented through a formal **Change Request** agreed upon by the Client and Capstone Student Project Team and, where required, approved by the relevant Carnegie Mellon University faculty advisor or academic program representative.

7. Academic Disclaimer and Sign-Off

As this is an academic capstone project, this document is intended solely for educational purposes and is not a legally binding commercial contract.

The service, architecture, benchmark, results, analyses, and recommendations developed through this project are academic deliverables and should not be interpreted as commercially warranted production systems or professional security certifications.

By signing below, both the Client and the Capstone Student Project Team acknowledge these terms and accept the scope, deliverables, timeline, assumptions, exclusions, and academic nature outlined in this Scope of Work document.

For the Client:

Approved by: _____

Date: _____

For the Capstone Student Project Team:

Approved by: Zhexuan Ye, Anmol Sahu, Yudi Guo, Karina Yang, Mark Chu

Date: _____